THE REMIX

LaLonde, Dehejia–Wahba, and Why the 2×2 Still Matters

The story behind yesterday's homework

Scott Cunningham · Baylor University

University of Glasgow · May 19, 2026

Causal Inference: The Remix · Yale University Press · forthcoming 2026

LESSON

1. LaLonde 1986

The National Supported Work Demonstration

A federally funded **randomized** job-training program.

1975–1979. Four target groups, including disadvantaged adult men.

Random assignment to treatment vs. control. Outcome: real earnings in 1978.

LaLonde's experimental benchmark

$$\widehat{\mathrm{ATT}}^{\mathrm{RCT}} \approx \$886$$

1978 dollars. Adult-male subsample.

Random assignment $\Rightarrow$ no selection. This is the truth.

Then LaLonde threw away the controls

He kept the treated — the 297 NSW men.

He replaced the controls with samples from the **CPS** and the **PSID**.

A simulation of what an applied researcher would face — no experiment, just observational data and methods.

Then he tried every method going at the time

- OLS with controls
- Two-step Heckman selection corrections
- Fixed effects on pre-program earnings
- Difference-in-differences

Could any of them recover $886?

No

Estimates ranged from $-$**\$16,000** to $+$**\$3,000**.

Most were negative. Most missed the benchmark by orders of magnitude.

LaLonde (1986), AER 76(4): 604–620.

The headline that shook applied econometrics

"The econometric procedures cannot replicate the experimental results."

This is the paper that made program evaluation move toward randomization and design.

LESSON

2. Dehejia & Wahba 2002

Dehejia & Wahba reopened the case

Rajeev Dehejia and **Sadek Wahba**,
Review of Economics and Statistics (2002).

They asked: does **propensity score matching** do better than LaLonde's methods?

First, they re-cut the experimental sample

They restricted to NSW men with earnings observed in **both 1974 and 1975**.

A smaller, slightly different population.

This is the subsample the homework uses: `lalonde_exp_panel.dta`.

A new experimental benchmark

$$\widehat{\mathrm{ATT}}^{\mathrm{RCT}}_{\mathrm{D\&W}} \approx \$1{,}794$$

Roughly **$1,700** in the way we usually report it.

Why bigger than LaLonde's $886? Different subsample, different ATT.

Why the headlines differ

- Both numbers come from the *same* NSW experiment.
- LaLonde used the full adult-male sample.
- D&W used a *subset* — only men with two years of pre-program earnings.
- Subsetting on a covariate changes the population whose ATT you're estimating.
- Same identification (random assignment). Different target group.

This is not a contradiction — it is the ATT defined on different units.

D&W's quasi-experimental result

With **propensity-score matching** on CPS / PSID controls...

they recovered $\approx \$1{,}700$.

A constructive answer to LaLonde: design plus matching can work.

LESSON

3. What you did yesterday

Yesterday's lab was on D&W's experimental subsample

`lalonde_exp_panel.dta` — a panel of NSW men, years 74, 75, 78.

The benchmark (post-period diff): $\$1{,}794$.

You had no covariates. Just `re`, `ever_treated`, `year`. $n_T = 185$, $n_C = 260$.

Step 1: four averages, by hand

	Pre (75)	**Post (78)**
Treated	$\bar{Y}^T_{75}$	$\bar{Y}^T_{78}$
Control	$\bar{Y}^C_{75}$	$\bar{Y}^C_{78}$

Four averages. Three subtractions. One DiD.

Step 2: the DiD

$$\widehat{\delta}_{\text{DiD}} = (\bar{Y}^T_{78} - \bar{Y}^T_{75}) - (\bar{Y}^C_{78} - \bar{Y}^C_{75})$$

$$= (\$6{,}349 - \$1{,}532) - (\$4{,}555 - \$1{,}267) = \$4{,}817 - \$3{,}288$$

$$\widehat{\delta}_{\text{DiD}} = \$1{,}529$$

Post-period diff-in-means alone: $\$6{,}349 - \$4{,}555 = \$1{,}794$.

Two valid estimators, two different numbers

Post-period diff-in-means (78 only)	\$1,794
DiD (78 vs. 75)	\$1,529
Difference	\$ 265

The pre-period gap was $\bar{Y}_{75}^{T} - \bar{Y}_{75}^{C} = +\$265$.

DiD subtracts it. Post-period diff doesn't.

Unbiasedness is a repeated-sampling concept

Random assignment makes $\mathbb{E}[\widehat{\delta}] = \text{ATT}$

across hypothetical re-randomizations of the experiment.

It does *not* make $\widehat{\delta} = \text{ATT}$ in any one sample.

Our pre-period happened to have $+\$265$ of random imbalance. Different draw, different imbalance, different DiD.

Step 3: three regressions, same number

- **OLS with interactions**: `re ~ ever_treated * post`
- **TWFE**: `feols(re ~ ever_treated:post | id + year)`
- **Long differences**: `lm(re78 - re75 ~ ever_treated)`

All three return $\widehat{\delta} = \$1{,}529$, to machine precision.

The equivalence breaks once we add staggered timing, covariates, or weights. Tomorrow.

Step 4: the event study tells the same story

- $\beta_{74} = -\$277$ — small, close to zero. **Pre-trend check passes.**
- β_{75} is the omitted baseline (zero by construction).
- $\beta_{78} = +\$1{,}529$ — same as the DiD.

Each event-study coefficient is a 2×2 DiD with 75 as baseline. Same arithmetic, three points.

LESSON

4. Why the ATT didn’t move

A puzzle

We used **non-experimental controls** (CPS, PSID) in the larger LaLonde–D&W literature.

And yet the ATT we target is still $\approx \$1{,}700$.

Why doesn't the ATT change when we change the controls?

Because the ATT is a property of the treated

$$\text{ATT} \;=\; \mathbb{E}[Y^1 - Y^0 \mid D = 1,\ \text{Post}]$$

Only the $D = 1$ side appears.

The control units are just a *tool* to learn about $\mathbb{E}[Y^0 \mid D{=}1, \text{Post}]$.

Random assignment gave us a clean tool

Under random assignment:

$$\mathbb{E}[Y^0 \mid D{=}1] \;=\; \mathbb{E}[Y^0 \mid D{=}0]$$

So $\mathbb{E}[Y \mid D{=}0, \text{Post}]$ *is* the counterfactual we need.

No selection bias.

Non-experimental controls break that equality

CPS men and NSW participants are not interchangeable.

$$\mathbb{E}[Y^0 \mid D{=}1] \;\neq\; \mathbb{E}[Y^0 \mid D{=}0]$$

Selection bias enters.

NSW men had worse pre-program earnings. The CPS pool didn't.

But the *target* hasn't changed

$\mathbb{E}[Y^0 \mid D{=}1, \text{Post}]$ is a feature of the NSW men.

It does not depend on who we picked as controls.

What the controls control is *our estimate* of that target —
and the size of the bias.

So what is DiD for, in this story?

- In the RCT sample: DiD just *confirms* the post-period difference-in-means.
- In the non-experimental sample: DiD tries to *net out* the pre-existing level gap between NSW and CPS men.
- The truth is fixed at $1,700. The question is whether DiD's parallel-trends assumption is enough to recover it.

Did it?

In yesterday's experimental lab: yes (in expectation)

$$\widehat{\delta}_{\text{DiD}} = \$1{,}529 \quad \text{vs.} \quad \text{benchmark} \ \$1{,}794$$

Within sampling variation. Pre-trend was flat.

Design did its job.

This afternoon, with covariates and non-experimental controls — a harder test.

LESSON

5. Now make it harder

Throw away the experimental controls

Keep the 297 NSW treated men.

Replace the controls with a random sample of American households from the **CPS**.

Data: lalonde_nonexp_panel.dta. Same treated rows. New controls.

Selection is severe

	1974	1975	1978
NSW (treated, $n = 185$)	\$2,096	\$1,532	\$6,349
CPS (control, $n = 15{,}992$)	\$14,017	\$13,651	\$14,847
Diff	$-\$11{,}921$	$-\$12{,}119$	$-\$8{,}498$

$$\mathbb{E}[Y^0 \mid D{=}1] \ll \mathbb{E}[Y^0 \mid D{=}0]$$

The level gap is huge — and so is the growth-rate gap. DiD doesn't fix that.

Naive DiD on this sample

$$\widehat{\delta}_{\text{naive DiD}} = +\$3{,}621$$

Right sign. *Wrong magnitude.* The benchmark is $\$1{,}794$.

NSW men had unusually low pre-earnings ($\$1{,}532$) and bounced back fast.
CPS controls didn't. DiD attributes the gap in growth rates entirely to treatment.

Lesson

6. Four estimators to rescue the ATT

The roster

- **DiD with additive X** (your published spec) — one slope per covariate
- **Outcome Regression** (Heckman, Ichimura & Todd 1997) — model the trend on controls; impute the treated counterfactual
- **Inverse Propensity Weighting** (Abadie 2005) — reweight controls to look like the treated
- **Doubly Robust DiD** (Sant'Anna & Zhao 2020) — OR + IPW combined

Each lives or dies on a different assumption. Compare to the $1,794 benchmark.

DiD with additive X

```
reg re ever_treated##post X1 X2 ... Xk, robust
```

- Pooled OLS on all years. No fixed effects.
- Covariates enter *linearly* with one slope — same pre and post.
- Identifies the ATT only if Assumptions 1–6 hold (parallel trends, no anticipation, SUTVA, homogeneous τ in X, parallel X-trends for treated and control).

This is what `lalonde-did-stata.do` actually computes on the non-experimental panel.

Outcome Regression — impute the counterfactual

Use the *controls only* to model the trend:

$$\widehat{\mu}_0(X) \;=\; \widehat{\mathbb{E}}[\Delta Y \mid D{=}0,\; X]$$

Then impute the counterfactual change for each treated unit, and average:

$$\widehat{\delta}_{\text{OR}} \;=\; \frac{1}{N_1} \sum_{i:\, D_i=1} \big(\Delta Y_i - \widehat{\mu}_0(X_i)\big)$$

Identification rests on $\widehat{\mu}_0(X)$ being correctly specified.

Inverse Propensity Weighting

Estimate $\widehat{p}(X) = \Pr(D{=}1 \mid X)$ by logit. Weight controls by:

$$w_i \;=\; \frac{\widehat{p}(X_i)}{1 - \widehat{p}(X_i)}$$

$$\widehat{\delta}_{\text{IPW}} \;=\; \mathbb{E}[\Delta Y \mid D{=}1] \;-\; \mathbb{E}_w[\Delta Y \mid D{=}0]$$

Identification rests on $\widehat{p}(X)$ being correctly specified.

Doubly Robust DiD

If *either* $\widehat{\mu}_0(X)$ *or* $\widehat{p}(X)$ is correct, DR is consistent.

Use the pscore as a **weight,** not as a **regressor.**

Sant'Anna & Zhao (2020). Implemented in `drdid` (Stata, R).

Lesson

7. How close did each one get?

The comparison

Estimator (Scott's published `drdid all`)	**ATT**	**Distance from $1,794**
Naive DiD (no X)	$+\$3{,}621$	overshoots by $1,827
DiD with additive X (your `reg` spec)	$+\$3{,}522$	overshoots by $1,728
Design-aware estimators (`drdid all`):		
OR — Regression-adjusted (`reg`)	$+\$1{,}770$	$-\$24$
IPW — Abadie (`ipw`)	$+\$1{,}861$	$+\$67$
DR-IPW (`dripw`)	$+\$1{,}979$	$+\$185$
DR-improved (`drimp`)	$+\$2{,}033$	$+\$239$
RCT benchmark (post-period diff)	$+\$1{,}794$	—

Naive DiD and additive-X DiD both overshoot by $\sim \$1{,}800$ — additive controls barely help.
The `drdid` design-aware estimators all land within $\$240$ of the benchmark; OR is closest at $24 below.

OR lands closest; all four design-aware estimators are within $240

- **OR** (`reg`, $1,770) sits **$24 below** — closest of all.
- **IPW** (Abadie, $1,861) sits $67 above.
- **DR-IPW** ($1,979) sits $185 above.
- **DR-improved** ($2,033) sits $239 above — furthest of the design-aware row.
- Additive-X DiD ($3,522) barely moves off naive DiD ($3,621).

"Doubly robust" means robust to one misspecification, not magically best in every sample. With $n_T = 185$ and $n_C \approx 16{,}000$, sampling variation alone explains the spread among design-aware estimators.

What this lab teaches you

- The **ATT is fixed** ($\approx +\$1{,}794$). The controls only change the bias.
- **Additive X barely helps:** \$3,522 vs naive \$3,621. You need a design-aware estimator.
- OR, IPW, DR-IPW, DR-improved offer four different uses of the same X — all land within \$240 of the benchmark.
- On this dataset, **OR (regression-adjusted) lands closest** at \$24 below.
- No method is magic. Each rests on an assumption you can name.

The ATT lives on the **treated.**

The controls only tell you

whether your method
can **see it.**